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Unit – 5 : Session – 8 : SLO – 1 | ARM Pipeline, Registers and Instructions

■ I. Fill in the Blanks

1. Fetch — a 3-stage pipeline is Fetch, Decode, Execute.
2. Stack Pointer (SP) — R13 points to the top of the stack.
3. BL (Branch with Link) — calls a subroutine, saving the return address in R14 (LR).

■ II. True or False

1. TRUE — pipelining overlaps instruction stages in ARM processors.
2. FALSE — R14 is the Link Register; R15 is the Program Counter.
3. TRUE — ADDEQ executes the ADD only when the Z (equal) condition flag is set.

■ III. Match the Following

1. R13 — 3. Stack Pointer.
2. R14 — 1. Link Register.
3. LDR — 2. Load instruction.

■ IV. Numerical Answers

1. 3 — three pipeline stages (Fetch, Decode, Execute).
2. 2 — one LDR plus one STR per iteration = 2 memory operations.
3. 13 — R0 through R12 are the general-purpose registers.